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 Studierichting: Informatica
 Oefeningenreeks 4A nr. 31.14

$$I = \int \frac{x^2 \sqrt{x}}{1 + \sqrt{x}} dx$$

$$\text{Stel: } t = \sqrt{x} \Leftrightarrow x = t^2$$

$$\Rightarrow dt = \frac{1}{2\sqrt{x}} dx \Leftrightarrow dx = 2\sqrt{x} dt = 2t dt$$

$$= \int \frac{t^4 \cdot t^1 \cdot 2t}{t+1} dt = 2 \int \frac{t^6}{t+1} dt$$

$\begin{array}{r} t^6 \\ -t^6 \quad -t^5 \\ \hline 0 \quad -t^5 \\ \quad +t^5 \quad +t^4 \\ \quad \hline \quad 0 \quad t^4 \\ \quad \quad -t^4 \quad -t^3 \\ \quad \quad \hline \quad \quad 0 \quad -t^3 \\ \quad \quad \quad +t^3 \quad +t^2 \\ \quad \quad \quad \hline \quad \quad \quad 0 \quad t^2 \\ \quad \quad \quad \quad -t^2 \quad -t \\ \quad \quad \quad \quad \hline \quad \quad \quad \quad 0 \quad -t \\ \quad \quad \quad \quad \quad +t \quad +1 \\ \quad \quad \quad \quad \quad \hline \quad \quad \quad \quad 0 \quad +1 \end{array}$	$\begin{array}{r} t \quad +1 \\ \hline t^5 \quad -t^4 \quad +t^3 \quad -t^2 \quad +t \quad -1 \end{array}$
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$$= 2 \left[\int t^5 - t^4 + t^3 - t^2 + t - 1 dt + \int \frac{1}{t+1} dt \right]$$

$$\text{Stel: } y = t + 1 \Rightarrow dy = dt$$

$$= 2 \left[\frac{t^6}{6} + c - \frac{t^5}{5} + c + \frac{t^4}{4} + c - \frac{t^3}{3} + c + \frac{t^2}{2} + c - t + c + \int \frac{1}{y} dy \right]$$

$$= 2 \left[\frac{t^6}{6} - \frac{t^5}{5} + \frac{t^4}{4} - \frac{t^3}{3} + \frac{t^2}{2} - t + \ln |t+1| + c \right]$$

$$= \frac{x^3}{3} - \frac{2x\sqrt{x}}{5} + \frac{x^2}{2} - \frac{2x\sqrt{x}}{3} + x - 2\sqrt{x} + \ln |\sqrt{x} + 1| + c$$

References